AGRI-NOVA AI PROJECT

TEAM: AGRINOVA AI RWANDA (INNOCENT BIZIMANA)

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#TURNING AGRICULTURE RWANDA WITH TECHNOLOGY

PROJECT NAME: AGRINOVA AI

AI-Powered Smart Agriculture Assistant for Rwanda

1. PROJECT OVERVIEW (DESCRIPTION)

AgriNova AI is an intelligent agriculture technology platform designed to support farmers in Rwanda by providing reliable, accessible, and AI-powered agricultural information and recommendations in Kinyarwanda.

The platform aims to solve real agricultural challenges by connecting Artificial Intelligence, agricultural knowledge, weather information, market insights, and data-driven decision-making.

PROBLEM STATEMENT

1. Problems and Solutions

Identified problem	AgriNova AI solution
1.Declining wheat production (-10.56%)	a.AI-powered crop management advice and productivity recommendations.
2.Declining sorghum production (-10.45%)	b. Smart farming guidance, improved crop practices and weather-based advice.(data from metheo Rwanda)
3.Low access to improved seeds (18%)	c. Information about improved seed varieties and where to access them.
4.Crop diseases and pests	d.AI crop disease identification through uploaded photos.
5.Unpredictable weather conditions	e. Weather-based planting, irrigation and harvesting recommendations.
6.Post-harvest losses	f. Smart harvesting, processing and storage guidance.
7.Limited market information	g. Verified market prices and market intelligence.
8.Unequal crop productivity	h. Yield analysis and recommendations for improved farming practices.

Post-Harvest Management: AgriNova AI offers guidance on harvesting, handling, processing, and storage techniques to minimize agricultural losses.

2. MAIN OBJECTIVE

To empower Rwandan farmers with accessible AI technology that helps them:

- Identify possible crop diseases and pests.
- Understand agricultural problems and receive practical recommendations.
- Access weather-related agricultural advice.
- Improve crop productivity.
- Make better farming decisions.
- Reduce the time spent searching for agricultural information or contacting extension officers.

  main objective: to help people to access limitless informations about agriculture in Rwanda as given in the surveys and recorded researches

3. CORE FEATURES

A. AI Agricultural Assistant

Build an intelligent chatbot that communicates in Kinyarwanda.

Farmers should be able to ask questions such as:

- Kuki amababi y'ibigori ahinduka umuhondo?
- Nakora iki kugira ngo ibishyimbo bitangirika?
- Ni iyihe fumbire nakoresha mu buhinzi bw'ibirayi?

Targets people

- ✓ **Small Farmers**

The AI capacity:

1. Understanding of the problem.
2. Possible causes.
3. Recommended actions.
4. Preventive measures.
5. Warnings when expert verification is required.

B. Crop Disease Identification

Allow farmers to upload crop photographs.

The AI should:

- Analyze uploaded images.
- Identify possible symptoms.
- Suggest possible diseases or pest problems.
- Provide practical recommendations.
- Clearly communicate uncertainty.
- Recommend contacting agricultural experts when necessary.

C. Weather-Based Agricultural Intelligence

Integrate weather information to provide:

- Rainfall forecasts.
- Temperature information.
- Drought warnings.
- Farming recommendations based on weather conditions.
- Advice on planting and irrigation.

D. Agricultural Market Intelligence

Help farmers understand:

- Crop market prices.
- Potential markets.
- Market opportunities.
- Supply and demand trends.
- Agricultural value addition opportunities.

Never invent market prices. Use verified and updated sources.

E. Agricultural Knowledge Database

Develop a reliable knowledge base using credible agricultural sources, including:

- Rwanda Agriculture and Animal Resources Development Board (RAB).
- National Institute of Statistics of Rwanda (NISR).
- Ministry of Agriculture and Animal Resources (MINAGRI).
- FAO.
- Other verified agricultural research institutions In rwanda.

Clearly distinguish verified information from AI-generated suggestions.

4. TECHNOLOGY STACK

Use practical technologies that can operate within a limited budget.

- Python for backend development.

- Gradio or another suitable web framework.
- Google Gemini API free tier for AI assistance.
- JSON for the initial agricultural knowledge database.
- NISR open datasets for agricultural statistics.
- Weather APIs with free access where available.
- GitHub for source-code management.
- Render for web application hosting.

Keep the application modular and easy to maintain.

5. USER INTERFACE DESIGN

Create a modern, professional, responsive interface.

Requirements:

- Mobile-friendly and desktop-friendly.
- Entirely in Kinyarwanda for farmer-facing content.
- Professional agricultural color palette using green, white and subtle natural colors.
- AgriNova AI logo and agricultural identity.
- Simple navigation.
- Easy-to-use chatbot.
- Clear buttons and readable typography.
- Attractive dashboard.
- Accessible design for users with limited digital skills.

6. AI QUALITY AND SAFETY

7.

8. It Will Help the Farmers to Access on Information from Survey Recorded and Research, When A Famer Ask Our Ai, Any Question It Will Gives Information About This Crops from NISR And RAB And Give Him, Her Documentation Explaining It Well

The AI must:

- Avoid presenting guesses as established facts.
- Never fabricate NISR statistics or RAB recommendations.
- Identify when information is uncertain.
- Avoid dangerous pesticide or fertilizer recommendations.
- Encourage consultation with qualified agronomists for serious crop problems.
- Provide clear, understandable Kinyarwanda responses.
- Explain the reasons behind recommendations.
- Our uniqueness is our Ai used only data from NISR and RAB

#Farmers don’t need promises ,they need Implementation.